Deployment Readiness Report: Town of Whitby Resident Support Assistant"

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- **GitHub link:** https://github.com/Srujanx/fraudguard-course-project/tree/main/docs/assignment1

AI-use statement. I used Claude (Anthropic) as a writing and planning assistant. It explained the assignment requirements, suggested the report structure, reviewed and corrected my drafts, and drafted sample paragraphs that I edited. I chose the use case and the Town of Whitby context, made the scoring and recommendation decisions, checked all prices in the Azure Pricing Calculator, and verified every source and service name myself. I take responsibility for the final content.

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1. Use Case and Readiness Snapshot

1.1 Use Case Description

The Town of Whitby is a Municipality for over 140,000 residents. The town will deploy a townwide resident support chat assistant to answer bylaw, garbage removal, service and development related queries for them to find the right support, assistance and escalate to a human if required and available. This will be available on the Town's official website, and a resident can chat, upload images and share their live location to interact with the agent. The agent reads a question and looks up its knowledge base which will contain Town's official documents and Support notes. Using RAG, the models ground every answer and cites the source. It will use gpt-4.1-mini LLM as it provides quick responses at cheap rates with acceptable hallucination levels. No custom model is trained for the first release. A terms and conditions and privacy policy will be prompted for the user to read and agree to use the chatbot. AI lifecycle stage that matters the most is Monitoring, simply because

resident's trust is the most important element for a Municipality. Hallucinated and misleading answers must be identified and fixed. Performance evaluations, drift detections, and cost monitoring run continuously alongside deployment and development.

1.2 Why Production AI

It should be a Production AI and not a prototype as it is a public facing chatbot that runs 24/7. During elections, storms and snow events there may be a spike in engagement and may demand automated scaling for such scenarios. Security and guardrails must be handled for instances like prompt injections and rate-limiting to avoid being manipulated or hacked. Every answer need logging, monitoring and audits must be conducted often to ensure reliability and accountability as the chatbot represents the Municipality.

1.3 Readiness Scores

Dimension	Evidence	Score
Data	exists on whitby.ca, but some documentation needs to review and updated with current information.	4
Model	RAG is proven algorithm which works effectively after training	4
Infrastructure	Town uses Azure Dev Tools, Containerizer and Cloud Platform for development	4
Organizational	Small IT Team with minimal specializations with system design, scale and optimizations	2
Risk	No privacy impact assessment yet, and no approved policy for public AI answers.	2
Cost	FinOps for tokens and Azure AI Licenses cost resealable amount of budget. Will also need emergency funds for court	3

	cases and fines in case of failure.	

1.4 Two Weakest Dimensions

The two lowest scores are organizational (2) and risk (2). The Town has only 37 IT staff working on multiple projects, so special attention to the chatbot cannot be promised. Minimal expertise of staff in MLOps and Production Applications as no dedicated role exists. Scaling or Security failure of the chatbot when associated staff is not available or on a vacation becomes a real problem as availability of equally capable role and skillset is hard to find to solve the problem. Before launching, the Town must name a business owner to review answer quality, name a technical owner in IT with a trained backup, and train staff on the human-handoff process. Another pillar which seems weak is risk. No privacy impact assessment and policy public approvals were conducted to avoid potential risks. Wrong or mislead answers can make a direct impact on public trust, and some information can lead to legal cases. This can impact stakeholder relationships, staff time, and cost to the Municipality. Before launching, the privacy impact assessment must be signed off, the AI-use policy approved, and content filtering and prompt-injection testing completed.

2. Deployment Environment Evaluation

2.1 Environment Fit Scorecard

Question	Score	Evidence
Data sensitivity	3	Personal information (names, locations, photos) is covered by MFIPPA.
Cost pattern	3	Apart from Azure Licenses, token costs are paid per use and are cheap. May spike during uncommon events.
Operational capacity	5	A small team with no MLOps role can't run GPU servers 24/7.
Latency	1	A chat reply within a few seconds is acceptable. The Model used is quick and on-prem servers are scalable

Applying decision rules – Hard Limiting: Only Latency hit 1 which forces to use on-prem GPU servers and Virtual Machines instead of cloud. Spread: Highest minus the lowest: 5 –1

= 4. So, the Deployment leans towards cloud. Average: (3+3+5+1)/4 = 3. This leads to a hybrid model.

2.2 Environment Comparison

One major policy for every municipality is the resident data should stay within Canadian borders and only Microsoft has their datacenters set up in Toronto. GCP and AWS are ruled out of the situation.

Criterion	Cloud	On premises	Hybrid
Privacy and residency	Data stays in Canadian datacenters if Azure is used.	Data never leaves the town, but the town does not have infrastructure and security expertise for this	Personal data stays within the town, and clouds are used for public data. This method is not cost efficient.
Handling spikes	Scales automatically	Risk of slowing down or failing if not optimized well.	The cloud and on-prem have different scaling rates. What to use when must justified
Upfront vs. ongoing cost	Monthly usage costs depend on use.	Large hardware purchase up front, plus power, cooling and replacement around year 4–5.	Highest cost total
Ops burden	Low. Microsoft offers support if tickets are raised.	High. The Town must patch, monitor, and take responsibility if anything breaks	Highest. Two environments to run and connect.
Model quality	Current models through Azure OpenAI, with built-in content filters.	Open-source models only. The Town must evaluate, update, and filter them.	Same as clouds.
Fit with existing systems	Use the Town's existing Microsoft 365 and Entra ID sign-in.	Separate stacks for staff to learn.	Needs secure links between both sides.

Microsoft Edge deployment was not considered, because the assistant runs in residents' web browsers and has no requirement for it.

2.3 Recommendation

Azure Cloud Services is something the town uses for all other projects and will consider the same for this one as it has data residency and is cost efficient. Only one question hit a hard score of 1. The scores for the group sit in a band of 2, which suggests the answers are aligned. The mean score is 3, which points to a mixed choice, but cloud is more beneficial in our case. Two items carry the decision the most. First, costs. Demand is low for most days. Then it jumps fast during storms and snow events. There are also spikes during changes to collection schedules and during election periods. Because of that pattern, pay-as-you-go cloud capacity costs less than keeping an on-premises GPU server built for peak loads. A peak-sized server would be unused most of the year. Second, staffing and operations. The Town has 37 IT staff. There is no dedicated MLOps position. That makes it hard to keep AI servers running, updated, and always scaled. Managed Azure services take on more of that ongoing work under Microsoft operations. Data sensitivity still matters. On premises was the strongest claim on that point. But MFIPPA does not ban cloud hosting. If the system is run in a Canadian region, with encryption and strict access controls, it meets the residency expectation. Also, an on-premises setup does not fix the staffing gap. The Town's work-order system will not move. It will stay in place and connect to the assistant through a secure API. This is an integration step. It is not a hybrid deployment of the AI.

2.4 Three-Year Cost Comparison

Assumption	Value	Basis
Conversations per month	8,000 on average	About 55,000 households assuming roughly 1 in 7 uses it each month
Turns per conversation	4	A question from the user and resolution in an ideal case. An escalation and feedback question in some cases.
Input tokens per turn	3,000	RAG adds about 2,000 tokens of document text
Output tokens per turn	300	A short answer with a citation
Monthly tokens	96M input, 9.6M output	8,000 × 4 × 3,000 and 8,000 × 4 × 300
Staff time, cloud	0.2 FTE	Monitoring and fixing bugs

Staff time, on-prem	0.5 FTE	patching, hardware, model updates
Loaded staff cost	$110,000 CAD per FTE per year	Check the Town's or Ontario's public-sector salary data
Exchange rate	1 USD = 1.37 CAD	Standard rate as of 8th Oct

Cloud line item	Rough monthly (USD)	How to price it
Azure OpenAI GPT-4.1 mini	~$55 at $0.40 / $1.60 per million tokens. Budget $100 for photos	96M × $0.40 + 9.6M × $1.60
Azure AI Search (Basic tier)	~$75	One Basic unit
Azure Container Apps	~$50	Consumption plan
Azure Cosmos DB (serverless)	~$25	Chat logs and requests
Blob Storage, Service Bus, Functions	~$15	Small amounts
Azure Monitor / Log Analytics	~$30	About 10 GB of logs a month
Private endpoints, Key Vault	~$30	About 4 private endpoints
Total	**~$325 USD (~$445 CAD) per month**	

Cost item (CAD)	Cloud, 3 years	On-prem, 3 years
GPU servers for failover, about $40k each	$0	$80,000
Cloud services ($445/month plus 30% spike buffer)	$21,000	$0
Power and cooling (about 3 kW, 24/7)	$0	$18,000
Hardware support contract	$0	$12,000
Staff time	$66,000 (0.2 FTE)	$165,000 (0.5 FTE)
One-time setup	$15,000	$25,000
Total	**≈ $102,000**	**≈ $300,000**

As of three years, cloud costs only about one third of On-prem. Staff time is the most important aspect and Cloud handles it for a reasonable price with Microsoft Support if required.

2.5 Tipping Point

The answer could be different if the council or the province says that residents' personal data must only live on Town servers and be handled there. In that case, the Data sensitivity part of the Environment Fit Scorecard would drop to 1. That score is a strict ceiling, so the plan would need an onsite or mixed setup, even if the rest of the score looks better. Cost by itself would not likely flip the decision. At roughly 1.7 cents per conversation, usage would have to rise about forty times. That would mean more than 300,000 conversations each month. Only then would token costs likely catch up to the on-premises option over three years. But on that same scale, an on-premises system would also need extra machines. So, the Town should keep an eye on any updates to MFIPPA. It should also follow new guidance from the Information and Privacy Commissioner of Ontario. The Town should watch for any new council direction on where data is kept. Since the application runs in containers and saves data in open formats, it could shift to Town servers. It could also run an open-source model if the rules change.

3. Cloud Platform and Service Selection

3.1 Platform Choice

Microsoft Azure was chosen for this use case for four main reasons. First, the city already uses Microsoft tools. It runs Microsoft 365, Microsoft Entra ID, and Windows Server. It also has an Azure subscription for backup. Because of that, identity setups, buying processes, and day to day work are familiar. Second, data residency matters here. Azure has Canada Central in Toronto and Canada East in Quebec City. Azure OpenAI Service also supports Canadian data zone options. This means prompts and data can stay in Canada, which is the key nonfunctional need noted in the risk view. Third, Azure fits the data and AI needs. Azure OpenAI Service, paired with Azure AI Search, gives a mature RAG setup. It is well documented and includes content safety features. Azure Machine Learning can also support later fine tuning, without forcing the city to adopt brand new tooling. Fourth, the cost and people fit the plan. Consumption pricing helps keep costs under control, and scale to zero supports a tight budget. The Microsoft public sector partner network is also larger than the others. Local hiring is easier because more candidates already know Microsoft platforms. AWS was considered, but it would bring extra overhead. It would add a second identity and billing system, plus a separate operations stack next to the city's Microsoft setup. That adds work but does not add new capabilities. Google Cloud also looks capable. It has

strong AI support, including Vertex AI, and it offers regions in Toronto and Montréal. Still, its Canadian municipal public sector ecosystem is smaller. It also matches the city's existing Microsoft agreements less well.

3.2 Service Map

Block	Azure service	What it does in this system	Model	Compute type
Front door / security	Azure Front Door with Web Application Firewall	Rate limiting and blocking of attack traffic.	PaaS	Managed edge
Serving (chat API)	Azure Container Apps	Runs the chat backend that receives questions, calls search and the LLM, and streams answers	PaaS	Container
Managed AI: LLM	Azure OpenAI (GPT-4.1 mini, regional Standard deployment in Canada)	Writes answers from the retrieved documents and reads uploaded photos	PaaS	Managed
Managed AI: retrieval	Azure AI Search	Indexes Town public pages and policies and returns the passages used for each answer (RAG)	PaaS	Managed
Managed AI: safety	Azure AI Content Safety (Prompt Shields)	Detects prompt injection and harmful content before and after the LLM	PaaS	Managed
Storage: documents	Azure Blob Storage	Stores the source documents that feed the search index	PaaS	—

Storage: conversations	Azure Cosmos DB (serverless)	Stores chat logs (with personal information masked) and service-request status	PaaS	Serverless
Integration	Azure Service Bus	Queues new service requests and delivers them to the Town's work-order system	PaaS	Managed
Notifications	Azure Functions	Sends residents status updates when a request changes	PaaS	Serverless
Evaluation (no training in v1)	Microsoft Foundry (evaluations)	Runs a test set of real resident questions before each release and scores accuracy and grounding	PaaS	Managed
Monitoring	Azure Monitor + Application Insights	Tracks answer quality signals, errors, response time and cost	PaaS	—
Secrets / identity	Azure Key Vault + Microsoft Entra ID	Stores keys. Controls which staff can access.	PaaS / SaaS	—

3.3 Containers vs. Serverless

Question	Chat API	Status notifications
Is Traffic steady or occasional?	Steady all day, spikes during storms and events	Occasional. It only runs when a work order changes status.

How long does each job run?	Long. It streams an answer word by word, calls search and the LLM and holds the conversation open.	Very short. It sends one message in about a second, then stops.
Does it need to move?	Yes	No. It's a few lines of code that can easily be rewritten
Result	**Container**	**Serverless**

The assistant picks a method by checking traffic, then how long it runs, then how portable it needs to be. For the chat part, the system starts as a container on Azure Container Apps. Traffic stays fairly even during the day. During storms, it jumps up fast. Because of that, at least one instance stays all the time, so people do not hit cold starts. When demand grows, it adds more instances by itself. Each chat call also takes time. It pulls documents, then it calls the LLM. After that, it sends the reply in a stream for over a few seconds. That pattern fits a container setup better than a function, since functions have hard time limits. The same container image can also run without changes on Azure Kubernetes Service, on another cloud, or on servers owned by the Town. For status notifications, the design uses Azure Functions. These events happen only from time to time, and they are triggered by specific changes. They finish quickly, in about a second. With serverless, the Town pays when a status change request occurs. The code size is small, so Azure-specific dependency is less of a concern.

3.4 Build vs. Buy

Option	What it means	Verdict
A. Buy everything (SaaS): Copilot Studio agent on the website	Low code. The team already uses it for other agentic projects. Fastest to launch.	Strong for a pilot, rejected for production.
B. Buy the model, build a thin layer: Azure OpenAI + AI Search + a small custom chat API	Microsoft provides the model and search. The Town builds only the code that connects them.	**Chosen**
C. Build everything: train or fine-tune an own model	Needs ML engineers, GPUs and training data the Town doesn't have	Rejected

We reviewed three paths. Option one was to train or fine tune a custom model. We did not pick that route. The Town does not have ML staff, no GPU setup, or no labelled training data. Also, a readymade model with RAG seems to cover what is needed. Option two was to build the assistant in Copilot Studio. Town staff already use it for internal work. That means it would be quicker to launch. It also fits a small internal pilot well.

We did not choose Copilot Studio for public service. The main issue is control. We have less control over how personal data is hidden, how photos are handled, how chats are stored, and where safety rules are placed. There is also a deployment limit. So, the recommended plan is different. We will buy the model and use the search. This would use Azure OpenAI and Azure AI Search. Then we would make only a thin chat API and host it in a container. We would start from Microsoft's open-source RAG reference app. That is faster than writing the API from the ground up.

3.5 Cloud Adoption Framework

STRATEGY: The goal is nonstop access to Town info. It should also cut down on repeat calls to Service Whitby. We will judge progress by how often questions get answered without handoff, how accurate the answers are on a set of test questions, how satisfied residents feel, and the cost to handle each chat.

Plan: The Town will pick a business lead from Customer Service. It will also assign a technical lead from IT. Each role will have a trained backup. Two IT staff members will finish Azure AI training. Customer Service will review public pages and written policies to confirm they are still current. Those items go into the knowledge base only after review.

Ready: IT will set up a separate Azure subscription for the AI work. This will act as a landing zone. Staff will get access using existing Entra ID roles. Azure Policy will limit where resources can run so they stay in Canadian regions. Private networking will be set up. Key Vault will be used. Budget alerts will be turned on at 50%, 80%, and 100% of the monthly budget. These steps will be taken before any resident data is stored.

Adopt: The assistant will be created using Microsoft's RAG reference app. It will start in one service area first. We will choose waste and recycling because it has a lot of questions, answers that are mainly factual, and low risk. We will test it with internal staff and then run a small public trial. After that, it will expand to by-law topics, Town services, and questions tied to development.

Govern: Azure Policy will enforce region rules, tagging rules, and security controls. The Town's AI use policy, and the privacy impact review will spell out what the assistant can say and what it is allowed to keep. The Town will review costs and a sample of conversations each month.

Secure: Microsoft Defender for Cloud will be used. Private endpoints will be added. Content Safety Prompt Shields will be enabled. Front Door's firewall will be set up to

protect the system. Prompt injection checks and penetration testing will be done before launch. Manage: Azure Monitor will notify the technical lead and the backup when there are errors, slow replies, or sudden cost increases. The evaluation test set will be run again.

3.6 Lock-in

The biggest lock-in risk is in the managed AI layer. Azure AI Search is the hardest piece to swap out. Its index format and ranking rules are tied to Azure. To reduce the risk, the Town keeps the source documents in Blob Storage as-is. The chunking logic also stays in the Town's code repo. With that setup, the index can be rebuilt in a non-Azure search system. Azure OpenAI is also somewhat hard to move. The prompts are tuned for one model. To handle that, there is one adapter module in front of model calls. The team also keeps an evaluation set. It is used to check any new replacement model. Chat logs are saved as plain JSON. They are exported on a set schedule. The chat API runs in a normal container. That container can be shifted to another cloud, or it can run on Town servers. Some lock-in is allowed on purpose.

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4. Risks and Decision

4.1 Top Three Risks

#	Risk	Likelihood	Impact	Mitigation (already in your design)	Owner
1	**Wrong or misleading answers.**	Medium	High	RAG over current public documents only, a citation on every answer, human handoff when confidence is low	Technical owner (IT)

2	**Privacy breach from misconfiguration.**	Low–Medium	High	Private endpoints, Entra ID roles, Key Vault, personal information masked in logs, Defender for Cloud, Azure Policy, and a privacy impact assessment before launch	Technical owner (IT)
3	**Cost spike or abuse.**	Medium	Medium	Front Door rate limiting, a per-conversation token cap, budget alerts at 50/80/100% (3.5), and a daily cost check during major events	Technical owner (IT)

4.2 Go / Delay / No-Go

I recommend Delay. The use case makes sense, and the cloud plan on Azure matches the Environment Fit result. The three-year total cost looks low, so we do not need to stop the plan. Still, two readiness areas look weak. Organizational is 2 out of 5, and Risk is 2 out of 5. Those gaps have to be closed before residents depend on the assistant. We should keep working on the landing zone, the building, and the pilot. The assistant should only go live to the public after all seven items below are met. Also, the Town should not launch during the municipal election window from October 26, 2026. If something goes wrong then, it will cause the most harm. It would also line up with the time when demand is at its peak.

5. Deployment Diagram

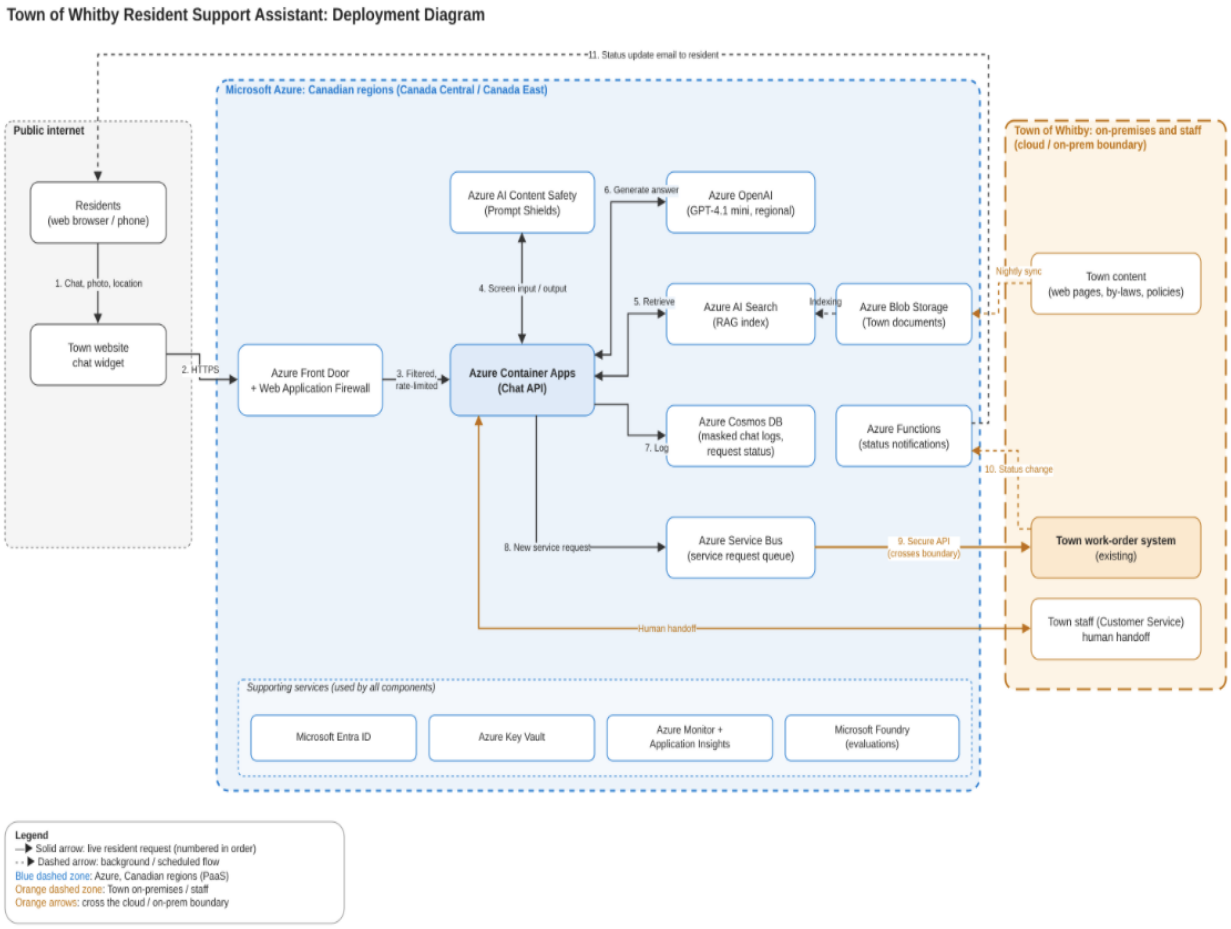


Figure gives the request path. A resident sends a message and first lands in Azure Front Door. That layer checks for attacks and enforces rate limits. After that, the message goes to the chat API running on Azure Container Apps. Next, the chat API reviews the message using Content Safety. It then looks up related text via Azure AI Search. After the lookup, it sends the question and the found text to Azure OpenAI. The system returns an answer with citations. At the same time, it writes a masked log entry to Cosmos DB.

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Appendix A: Readiness Scorecard

Scores use a 0–5 scale: 0–1 not ready, 2 major gaps, 3 workable with fixes, 4 mostly ready, 5 fully ready. A summary appears in Section 1.3; the two weakest dimensions are discussed in Section 1.4.

Dimension	Score	Evidence	What would raise the score
Data	3	Public service pages, by-laws and policies exist on the Town website, but some are outdated or available only as PDFs.	Content audit by Customer Service; outdated pages corrected before indexing.
Model	4	An off-the-shelf LLM (GPT-4.1 mini) with retrieval-augmented generation meets the need; no custom training is required.	A passed evaluation on 200 real resident questions.
Infrastructure	3	Microsoft 365 and Entra ID are in place, but there is no production Azure landing zone for AI workloads.	Landing zone with a Canada-only region policy and private networking (CAF Ready phase).

Organizational	2	The IT team of 37 staff supports many projects; no dedicated MLOps or AI operations role exists.	Named business and technical owners, a trained backup, and Azure AI training for two staff.
Risk / Governance	2	No privacy impact assessment has been completed and no policy on public AI answers has been approved.	PIA signed off, AI-use policy approved, prompt-injection testing passed.
Cost	3	Cost per conversation is low, but demand spikes during storms and elections are hard to forecast.	Budget alerts, token caps, and three months of real usage data.
Total	**17 / 30**	**Average 2.8: workable, with two dimensions that must be fixed before launch.**	

Table A1. Readiness Scorecard for the Town of Whitby resident support assistant.

Appendix B: Environment Fit Scorecard and Comparison

B.1 Environment Fit Scorecard

[Check against the template wording] Scale: 1 = hard limit that requires on-premises deployment; 5 = strong fit for cloud deployment.

Question	Score (1–5)	Evidence
Data sensitivity	3	Names, locations and photos are personal information under MFIPPA. Cloud use is permitted when data stays in Canada (Azure Canadian regions).
Cost pattern	5	Demand is quiet most days with sharp spikes during storms, snow events, collection-schedule changes and elections; pay-per-use pricing fits.
Ops capacity	5	A small IT team with no MLOps role cannot run and patch GPU servers 24/7.
Latency	4	A reply within a few seconds is acceptable; there is no offline or real-time requirement.
Average	**4.25**	**Spread 2 (5 − 3); no hard limits.**

Table B1. Environment Fit Scorecard.

Decision rule applied in order:

1. Hard limits: no question scored 1, so nothing forces on-premises deployment.
2. Spread: highest minus lowest score is 5 − 3 = 2, so the answers are consistent and do not indicate a mixed (hybrid) case.
3. Average: (3 + 5 + 5 + 4) ÷ 4 = 4.25, which points to cloud deployment.

B.2 Environment Comparison

Cloud: everything runs in Azure Canadian regions, with the model provided by Azure OpenAI. On-premises: a GPU server in the Town data centre runs an open-source LLM, operated by Town staff. Hybrid: the LLM runs in the cloud while personal data and chat logs stay on Town servers.

Criterion	Cloud	On-premises	Hybrid
Privacy and residency	Data stays in Canada. Microsoft secures the platform; the Town configures access.	Data never leaves Town servers. Strongest control, but the Town carries the full security load.	Personal data stays on Town servers, but every question still goes to the cloud LLM, so the privacy gain is small.
Handling spikes	Scales automatically during storms and elections; pay only for what is used.	A server sized for storm peaks sits mostly idle; one sized for normal days slows or fails during storms.	The cloud side scales, but on-premises logging and storage can become the bottleneck.
Upfront vs. ongoing cost	No hardware purchase; monthly usage cost.	Large hardware purchase, plus power, cooling and replacement around year 4–5.	Pays for both environments; highest total cost.
Ops burden	Low. Microsoft patches and scales; fits a 37-person team with no MLOps role.	High. The Town patches, monitors and fixes it 24/7.	Highest. Two environments to run and connect.
Model quality	Current models through Azure OpenAI, with built-in content filtering.	Open-source models only; the Town must evaluate, update and filter them itself.	Same as cloud.
Fit with existing systems	Uses existing Microsoft 365 and Entra ID sign-in.	Separate stack for staff to learn.	Needs secure links between both sides.

Table B2. Comparison of deployment environments for the resident support assistant.

Edge deployment was not considered because the assistant runs in residents' web browsers and has no offline or on-device requirement.

Appendix C: Service Map and Cost Table

C.1 Service Map

Block	Azure service	Role in this system	Model	Compute
Front door / security	Azure Front Door + Web Application Firewall	Public entry point for the chat; rate limiting and blocking of attack traffic.	PaaS	Managed edge
Serving (chat API)	Azure Container Apps	Runs the chat backend: receives questions, calls search and the LLM, streams answers.	PaaS	Container
Managed AI: LLM	Azure OpenAI (GPT-4.1 mini, regional Standard deployment in Canada)	Writes answers from retrieved documents and reads uploaded photos.	PaaS	Managed

Managed AI: retrieval	Azure AI Search	Indexes Town public pages and policies; returns passages for each answer (RAG).	PaaS	Managed
Managed AI: safety	Azure AI Content Safety (Prompt Shields)	Detects prompt injection and harmful content before and after the LLM.	PaaS	Managed
Storage: documents	Azure Blob Storage	Stores source documents that feed the search index.	PaaS	—
Storage: conversations	Azure Cosmos DB (serverless)	Stores chat logs (personal information masked) and service-request status.	PaaS	Serverless
Integration	Azure Service Bus	Queues new service requests and delivers them to the Town work-order system.	PaaS	Managed
Notifications	Azure Functions	Sends residents status updates when a request changes.	PaaS	Serverless
Evaluation (no training in v1)	Microsoft Foundry (evaluations)	Runs a test set of real resident questions before each release and scores accuracy and grounding.	PaaS	Managed
Monitoring	Azure Monitor + Application Insights	Tracks answer-quality signals, errors, response time and cost.	PaaS	—
Secrets / identity	Azure Key Vault + Microsoft Entra ID	Stores keys; controls which staff can access what.	PaaS / SaaS	—

Table C1. Azure service map. The Town work-order system remains outside Azure and receives requests through Service Bus.

C.2 Cost Assumptions

Assumption	Value	Basis
Conversations per month	8,000 average (about 5,000 normal; about 20,000 in storm or election months)	About 55,000 households; roughly 1 in 7 uses the assistant each month
Turns per conversation	4	Question, follow-up, clarification, then handoff or request
Input tokens per turn	3,000	Question + about 2,000 tokens of retrieved documents + chat history
Output tokens per turn	300	Short answer with a citation
Monthly tokens	96M input; 9.6M output	8,000 × 4 × 3,000 and 8,000 × 4 × 300

Staff time	Cloud 0.2 FTE; on-premises 0.5 FTE	On-premises adds patching, hardware, model updates and on-call
Loaded staff cost	$110,000 CAD per FTE per year	[Source for salary figure]
Exchange rate	1 USD = 1.37 CAD	Bank of Canada, accessed [date]

Table C2. Assumptions behind the three-year cost comparison.

C.3 Cloud Price List

Service	Tier / setting	Monthly (USD)	Source	Accessed
Azure OpenAI GPT-4.1 mini	Regional Standard, Canada; 96M input / 9.6M output tokens + photo buffer	~100	Azure OpenAI pricing page	[date]
Azure AI Search	Basic, 1 unit	~75	Azure Pricing Calculator	[date]
Azure Container Apps	Consumption, minimum 1 replica	~50	Azure Pricing Calculator	[date]
Azure Front Door	Standard + WAF rules	~35	Azure Pricing Calculator	[date]
Azure Monitor + Application Insights	About 10 GB logs per month	~30	Azure Pricing Calculator	[date]
Private endpoints + Key Vault	About 4 private endpoints	~30	Azure Pricing Calculator	[date]
Azure AI Content Safety	About 64,000 checks per month (input + output)	~25	Azure Pricing Calculator	[date]
Azure Cosmos DB	Serverless	~25	Azure Pricing Calculator	[date]
Blob Storage, Service Bus, Functions	Small usage	~15	Azure Pricing Calculator	[date]
Total		**~385 USD (~527 CAD)**		

Table C3. Estimated monthly cloud costs, priced in the Azure Pricing Calculator on the access dates shown.

C.4 On-Premises Price List

Item	Estimate (CAD)	Source	Accessed
2 × GPU servers for failover (e.g., NVIDIA L40S)	80,000	[Vendor list price page]	[date]
Power and cooling, 3 years (about 3 kW, 24/7)	18,000	[Ontario electricity rate source]	[date]
Hardware support contract, 3 years	12,000	[Vendor support pricing]	[date]

Table C4. Estimated on-premises hardware and facility costs.

C.5 Three-Year Total Cost (CAD)

Cost item	Cloud	On-premises
Hardware (2 GPU servers)	0	80,000
Cloud services (527 × 36 months × 1.3 spike buffer)	24,700	0
Power, cooling and hardware support	0	30,000
Staff time (0.2 vs. 0.5 FTE)	66,000	165,000
One-time setup	15,000	25,000
Total	**≈ 106,000**	**≈ 300,000**

Table C5. Three-year cost comparison.

Sensitivity: the most uncertain line is conversation volume and tokens per conversation. At five times the assumed volume (40,000 conversations a month), the Azure OpenAI cost rises from about $1,650 to about $8,200 CAD per year, an increase of about $6,500. The cloud option remains about one-third the cost of on-premises.